What is Genetics?: the concept of heredity

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Choose the correct word from this list to fill each space below. Each word has been used once:

Adapt	*cell*	*classical*	*colour*	*disease*	*DNA*
evolution	*expressed*	*genes*	*genetics*	*genome*	*genotype*
half	*inherited*	*Mendel*	*molecules*	*origins*	*pea*
pedigree	*phenotype*	*populations*	*selective*		

How often have you been told that you resemble a parent (and groaned as a result)? We now understand that many features are _____ from our parents and, in turn, from their parents and so on, as a result of each parent contributing _____ of their offspring's genetic make-up. Resemblances occur when one parent's contribution is _____ more strongly than the other parent's. These features can be as simple as eye or hair _____, while other features can have more serious consequences, such as the inheritance of _____.

It is important to understand the difference between the terms *phenotype* and *genotype*. _____ is used to describe observable features, such as hair colour, etc. The genetic make-up of an individual is its _____. This includes all genes inherited from both parents, **whether or not they are expressed**. Genetic material, _____, is stored in the nucleus of a _____.

There are three main branches of genetics: *Mendelian*, *molecular* and *population*. Mendelian genetics is also known as _____ genetics because it is the branch begun over a century ago by Gregor Mendel. _____ studied how certain features are transmitted from one individual, in the case of his studies the _____ plant, to its offspring. This form of _____ breeding is still used today when we want to produce, or enhance, a certain characteristic in a plant or animal. A common example is the selective breeding of farm animals for sale as meat, or for *purebred* (_____) stock. Because it is the study of transmission, Mendelian _____ is often called transmission genetics.

As the name suggests, molecular genetics studies genetics at the level of individual _____. Molecular geneticists study the composition of _____, how they are controlled and what they produce when they are all put together. The complete set of genetic information an organism contains is called its _____. The ability to produce genetic 'maps' is a result of advances made in molecular genetics.

Population genetics examines _____ as a whole, rather than individual organisms. Population geneticists might ask questions such as, '*What makes one group better able to* _____ *than another population of the same species?*', '*What are the differences and similarities among populations?*' and '*How has* _____ *directed the way populations have changed over time?*' Population genetics reveals relationships between species, often providing clues to the _____ of species.

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